MILITARY SPECIFICATION

DISTILLER SCALE PREVENTIVE TREATMENT FORMULATIONS (METRIC)

This amendment forms a part of DOD-D-24577(SH), dated 18 September 1978, and is approved for use within the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, after "SPECIFICATIONS, FEDERAL": Add:

"MILITARY
MIL-S-16216 - Steel Plate, Alloy, Structural, High-Yield Strength (HY-80 and HY-100).
MIL-S-22698 - Steel Plate: Carbon, Structural, For Ships."

2.2, under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)" add:

"A 213 - Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes.
"A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusions-Welded Unifired Pressure Vessels.
"B 167 - Nickel-Chromium-Iron Alloy (UNS N06600) Seamless Pipe and Tube.
"B 168 - Nickel-Chromium-Iron Alloy (UNS N06600) Plate, Sheet, and Strip.
"D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (And the Calculation of Dynamic Viscosity).
"G 1 - Preparing, Cleaning, and Evaluating Corrosion Test Specimens.
"G 38 - Making and Using the C-Ring Stress-Corrosion Cracking Test Specimen."

* Bottom of page, delete "FSC 4620" and substitute "FSC 6850".

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3.2, 3.2.1, 3.2.2, 3.2.3, and 3.2.4: Delete and substitute:

"3.2 Material. The distiller scale preventive treatment formulations shall be in aqueous medium and shall contain organic polyelectrolytes such as polyacrylates, and anti-foam agent which shall conform to the requirements specified herein.

"3.2.1 Unit content. The unit content shall be at least 6.0 gallons (or metric equivalent) of liquid at 25°C (77°F) as specified in 4.5.1.

"3.2.2 Maintenance of heat transfer coefficient. Using the recommended dosage of 10.5 ± 0.2 milligram per liter (mg/L), the distiller scale preventive formulation in the feedwater at the heat transfer coefficient shall be maintained at no less than 75 percent of the "clean tube value" throughout the 80-hour test period when tested in accordance with 4.6.1.

"3.2.3 Ease of scale removal. Scale formed on heat transfer surfaces, while using the distiller scale preventive treatment formulation, shall be greater than 99.0 percent removed by one acid cleaning using the procedure given in 4.6.1.2.

"3.2.4 Appearance. The distiller scale preventive treatment formulation shall be a clear liquid with no phase separation and shall form a homogeneous emulsion which is stable for 30 minutes when examined by the procedure given in 4.6.2."

Add as paragraphs 3.2.6.1, 3.2.6.2, and 3.2.8:

"3.2.6.1 General corrosion. The distiller scale preventive treatment shall not corrode 90-10 copper nickel (Cu-Ni), 70-30 Cu-Ni, copper, valve bronze, phosphor bronze, gun metal or K-Monel at a rate greater than 0.10 mm/yr (4.0 mil/yr) when specimens of these metals are immersed in 100 mg/L concentration of distiller scale preventive treatment formulation in distilled water and tested in accordance with 4.6.4.1.

"3.2.6.2 High temperature corrosion. Corrosion rates in excess of 0.04 mm/yr (1.5 mil/yr) for carbon steel, HY-80 steel, type 347/348 corrosion resistant steel (CRES), or Inconel Alloy 600; pitting in excess of 0.08 mm (3.0 mils) on Alloy 600 and type 347/348 steel and stress corrosion cracking on stressed C-rings of Inconel Alloy 600 and 347 CRES shall not occur when tested in accordance with 4.6.4.2.

"3.2.8 Homogeneity of an aqueous formulation solution. The distiller scale preventive treatment formulations shall form a stable homogeneous aqueous solution. A stable, homogeneous solution shall be defined as one which does not form a phase separation after standing for 24 hours at 25°C when tested in accordance with 4.6.6."

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3.4.2: Delete and substitute:

"3.4.2 Specific gravity. The density of the liquid shall not vary more than 0.01 grams per milliliter (g/mL) from the value obtained on qualification scale preventive. Specific gravity test shall be in accordance with 4.8.2."

PAGE 4

4.4.3: Delete and substitute:

"4.4.3 Sampling for tests. Three one-liter containers shall be filled with the distiller scale preventive formulation which shall be withdrawn from an agitated mixing tank prior to packing each lot. The samples shall be contained in plastic bottles such as polyethylene or other plastic containers which do not react with the chemicals of the distiller scale preventive treatment. The three equal portions of each sample shall be used as follows:

(a) Two for quality conformance tests (see 4.5.2).
(b) One for contractor.

The results of the tests required by 4.5.2 shall be submitted to NAVSEA or the designated laboratory."

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4.6.1 through 4.6.2: Delete and substitute:

"4.6.1 Maintenance of heat-transfer coefficient. The heat-transfer coefficient, "U", shall be determined by using a calibrated model submerged tube distiller such as the 38 liters per hour (L/h) (10 gallons per hour (gal/h)) unit schematically shown on figure 1. The heat-transfer coefficient after 10 hours of operation is designated as the "clean tube value." The "clean tube value" shall be greater than 2900 watts per meter^2 - kelvin, W/m^2 K (510) British thermal unit per hour - square foot-degree F, BTU/ft^2 h.°F). Test periods of distiller operation shall be conducted with synthetic seawater conforming to ASTM D 1141. The seawater shall be prepared by mixing sea salts in the proportion of 41 grams of salt per liter of water (5.5 ounces of salt per gallon), resulting in a 35,000 mg/L solution of sea salts. Seawater prepared in this manner shall have a pH of 8.1 $\pm$ 0.2. The distiller scale preventive treatment shall be added to the seawater in a concentration of 10.5 $\pm$ 0.2 mg/L. An additional 1 mg/L of polyethylene glycol monooleate (PEG 200-MO) anti-foam agent shall be added to the distiller feedwater to control excessive brine carryover in the test unit. (PEG-200-MO may be obtained from the ARMAK Co., State Road and Cottman Avenue, Philadelphia, Pennsylvania 19135 or equal.) The temperature data needed to calculate the heat transfer coefficient shall be recorded every hour during

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periods of operation. The distiller shall be operated at 85°C ± 1°C (185°F ± 2°F) boiling temperature, and the average feedwater flow rate shall be 1.9 liters per minute (L/min) (0.5 gallon per minute). The steam supplied by the boiler shall be saturated and at a temperature ranging between 100°C - 110°C (212°F - 230°F). The heat supplied by the boiler shall be constant at a rate of 30,000 W (102, 500 BTU/h). The brine concentration ratio shall be maintained at 1.5 ± 0.1. The test shall be conducted for 90 hours including the initial 10-hour break-in period.

"4.6.1.1 Percent heat-transfer retention calculations. The heat-transfer coefficient, "U", shall be expressed by the following equation:

$$U = Q/A\Delta T$$

ΔT (in units of °C or °F) is the apparent temperature difference between the saturated steam temperature and the boiling temperature in the evaporator chamber. "A" is the heat-transfer surface area in units of square meters (m^2) or square feet (ft^2), and Q is the heat flow rate in watts (W) or BTU/h. "Q" is determined from the electrical input (30,000 W) into the boiler. The heat-transfer coefficient, U, is affected by scaling of heat-transfer surfaces during operation of the distiller. After the first 10 hours of the test run, the heat-transfer coefficient shall be calculated by using the value of A for the distiller, the measured value ΔT and the "Q" value for the heat flow. The heat-transfer coefficient shall be calculated for every hour of operation. The percent heat-transfer retention shall be calculated as follows:

$$\text{percent heat-transfer retention} = \frac{U_2}{U_1} \times 100$$

U_1 = Clean tube heat-transfer coefficient W/m^2K determined after the first 10 hours of operation.
U_2 = Heat-transfer coefficient W/m^2K determined at 1-hour intervals during the last 80 hours of operation.

"4.6.1.2 Ease of scale removal. At the completion of the 90 hour test period, the ease of scale removal shall be determined by cleaning the distiller with 1 kilogram of inhibited sulfamic acid, dissolved in 20 kilograms of water, making a 5 percent weight solution. This solution shall be injected into the distiller and allowed to stand for one hour; the distiller shall be heated to 93°C (200°F) for another hour, and then rinsed with fresh water. A visual examination of the tubes shall be made to determine the effectiveness of the acid cleaning to descale the heat transfer surfaces. No more than 1.0 percent, 5800 mm^2 (9.0 in^2), of the 0.58 m^2 (6.25 ft^2) heat transfer surface area shall have scale remaining after the acid cleaning.

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"4.6.2 Appearance. The distiller scale preventive treatment formulation shall be visually examined. It shall be a clear liquid or it shall be a homogeneous, stable emulsion for 30 minutes after vigorously shaking a 125-mL sample for 1 minute in a stoppered 250-mL Erlenmeyer flask."

Add new paragraphs 4.6.4.1 through 4.6.4.2.4:

"4.6.4.1 General corrosion. Duplicate corrosion test coupons of 90-10 Cu-Ni, 70-30 Cu-Ni, Cu, Valve Bronze, Phosphor Bronze, Gun Metal and K-Monel shall be placed in separate standard taper one-liter Erlenmeyer flasks. The duplicate coupons shall be separated by Teflon washers. The coupons shall be submerged in 650 mL of solution in distilled water containing 100 mg/L of distiller scale preventive treatment formulation. A water-cooled, standard taper condenser shall be attached to the flask. The solution shall be heated and maintained at 99°C ± 1°C (210.2°F ± 2°F) for 96 hours. The coupon preparation and calculation of corrosion rate shall be conducted in accordance with procedure B of ASTM D 2688.

"4.6.4.2 High temperature corrosion test. The test C-ring specimen shall be prepared from seamless tubing of Inconel Alloy 600 and Type 347/348 CRES and the corrosion coupons shall be made from forged plates of carbon steel, HY-80 steel, Inconel Alloy 600, and Type 347/348 CRES. The tubing for the C-ring specimens shall meet requirements of ASTM B 167 for the Inconel Alloy 600 and requirements of ASTM A 213 for the Type 347/348 CRES. The specimens shall be 1.27 centimeters (1/2 inch) od with a minimum wall thickness of 0.135 centimeter (0.053 inch). Tubing specimens shall be fabricated from mill annealed tubing material which has been stress relieved at 607°C ± 14°C (1125°F ± 25°F) for 4 hours. The plate specimens shall meet requirements of MIL-S-22698 for carbon steel, MIL-S-16216 for HY-80, ASTM A 240 for Inconel Alloy 600, and ASTM B 168 for CRES 347/348. These mill annealed plate materials shall be tested in the stress relieved condition at 607°C ± 14°C (1125°F ± 25°F) for 4 hours per inch of thickness, (4 hours minimum).

"4.6.4.2.1 Mass change specimens. Two mass change coupons 25 mm x 51 mm x 3.2 mm (1 inch x 2 inch x 1/8 inch), of 347 CRES, Inconel Alloy 600, HY-80, and carbon steel shall be polished with 600 grit or finer metallurgical polishing paper, cleaned in alcohol or electronic grade acetone, dried in a desiccator, and weighed to the nearest 0.1 milligram.

"4.6.4.2.2 Stress corrosion cracking specimens. Stressed C-rings (2 each) of 347 CRES and Inconel Alloy 600 shall be fabricated from 1.25 centimeter od x 0.135 centimeter (1/2 inch od x 0.053 inch (minimum)) wall thickness, seamless tubing in accordance with ASTM G 38. The specimens shall be loaded beyond the proportional limit to have a maximum plastic strain of 5 to 10 percent. Stressed specimens shall receive no subsequent heat treatment.

"4.6.4.2.3 Pre-test examination. Fabricated specimens should receive a pre-test examination to ensure that no unacceptable pitting or cracking exists on the specimens prior to testing.

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"4.6.4.2.4 Test procedure. The weight change coupons shall be mounted in Teflon holders and these, along with the stressed C-rings, shall be suspended in a CRES static autoclave vessel. The latter shall be filled with enough aqueous solution containing 12,500 mg/L (p/m) distiller scale preventive treatment to assure at least a 25 mm layer of solution above all the specimens and to allow for expansion due to heating. The solution of distiller scale preventive treatment shall be analyzed and reported for pH, conductivity, chloride, and sodium before and after the test to determine if the solution changed during the testing. The sealed autoclave shall be heated to 307°C ± 5°C and maintained at this temperature for a continuous 720 hour period. After removing the specimens from the autoclave, each coupon shall be chemically cleaned and the corrosion rate of each mass change coupon shall measured in accordance with ASTM G 1 recommended practice. The applicable paragraphs for chemical cleaning (including Appendix A.2) shall be used for Inconel Alloy 600, 347 CRES, carbon steel, and HY-80 steel. Following the final chemical cleaning, each mass change coupon shall be rinsed in accordance with ASTM G 1, with either alcohol or electronic grade acetone, dried in a desiccator and then weighed to the nearest 0.1 mg. Corrosion rates shall be calculated using the equation, contained in section entitled "Calculations of Corrosion Rate", of ASTM G 1. C-rings shall be examined using a stereomicroscope at low magnification (about 50X) for stress corrosion cracks. The entire surface of each specimen shall be examined. In addition, all C-rings shall be sectioned transversely, mounted and metallographically examined for cracks at magnification up to 500X. No stress corrosion cracking shall be evidenced as determined by surface and metallographic examination."

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4.6.5: Delete and substitute:

"4.6.5 Compatibility with other treatment formulations. One hundred mL of a 10 percent solution of the distiller scale preventive treatment formulation shall be mixed at 25°C ± 1°C (77°F ± 2°F) with 100 mL of a 10 percent solution of each formulation previously qualified. The material being tested shall not be considered compatible if a percipitate forms when the two solutions are mixed."

Add new paragraph 4.6.6:

"4.6.6 Homogeneity of an aqueous formulation solution. Prepare a solution of the distiller scale preventive formulation by adding 7.5 mL of a well-mixed sample from a 10-mL graduated cylinder to a 500-mL volumetric flask. Add distilled water to the mark. Shake vigorously for 1 minute to homogenize the solution. Allow the solution to stand for 24 hours 25°C ± 2°C. No phase separation shall occur."

4.7.4: Delete and substitute:

"4.7.4 Polymer content - acetone-insoluble material. A 2 mL portion of the well-mixed liquid formulation at 25°C ± 1°C shall be measured by a volumetric pipette and transferred to a dried, preweighed 100 mL centrifuge tube plus a stirring rod. The centrifuge tube shall be weighed with an accuracy of plus or minus 0.2 milligrams. Twenty five mL of reagent grade acetone (water content, 0.5 percent maximum) shall be added to the sample and the mixture stirred with the stirring rod for 2 or 3 minutes. The stirring rod shall be

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removed from the tube and placed across the top of a beaker being careful not to loosen any of the adherent material. The tube shall be capped and centrifuged for 5 minutes at a speed at which the centrifuge is capable of producing a relative centrifugal force (ref) between 600 and 700 at the tips of the tubes. The supernatant liquid layer shall be removed from the polymer residue. A second 25-mL portion of acetone shall be added to the residue and the extraction procedure repeated. The acetone extraction shall be repeated with a third 25-mL portion of acetone. The solid polymer material remaining in the 100-mL centrifuge tube shall be gently warmed on a hot plate (surface temperature 60°C) in a well-ventilated hood to remove the bulk of the remaining acetone. The stirring rod shall be inserted into the tube, and the solid shall be dried in an oven at 105°C ± 2°C for 1 hour. The solid shall then be cooled in a desiccator over a suitable desiccant and the tube and stirring rod shall be weighed. The drying shall be repeated until successive weighings are within plus or minus 1 mg. The percentage of acetone insoluble polymer shall be within plus or minus 0.55 mass percent of the value obtained as specified in 3.3.4. The percentage of acetone insoluble shall be calculated as follows:

$$\text{Percent acetone insoluble material} = \frac{\text{mass of residue in centrifuge tube}}{\text{mass of original liquid sample}} \times 100"$$

4.8.2: Delete and substitute:

"4.8.2 Specific gravity. A clean, dry 25-mL pycnometer shall be weighed. Mass shall be recorded to the nearest 0.1 mg. The stopper shall be removed and the pycnometer filled with recently boiled distilled water that has been cooled. The stopper shall be inserted, any excess water removed, and the pycnometer allowed to reach 25°C. The pycnometer shall be weighed and the weight of the water recorded to the nearest 0.1 mg. The procedure shall be repeated by using the well-mixed liquid distiller treatment formulation. The specific gravity of the distiller treatment shall be calculated as follows:

$$\text{Specific gravity} = \frac{\text{mass of formulation - mass of pycnometer}}{\text{mass of water - mass of pycnometer}} ."$$

4.8.3: Delete and substitute:

"4.8.3 Intrinsic viscosity: The intrinsic viscosity shall be determined as follows:

(a) The intrinsic viscosity shall be determined at 25°C ± 0.1°C (77°F ± 0.2°F) by the dilution method using a Cannon-Ubbelohde four-bulb shear dilution viscometer with a shear range at the wall of 45 to 1800 sec^{-1}. The first step shall be to determine the reduced specific viscosity. For the necessary calculations of the reduced specific viscosity, a concentration of 100 grams per liter (g/L) for the active ingredient in the scale preventive formulation shall be assumed. Using a well-mixed sample, three dilutions shall be prepared by pipetting 10 mL (20 g/L), 7.5 mL (15 g/L) and 5 mL (10 g/L) into separate 50 mL volumetric flasks. Add 10 mL of 1 molar (M) sodium chloride solution to

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each flask and bring to the mark with distilled water to produce a final concentration of 0.2M sodium chloride. A fourth flask containing 0.2M sodium chloride and no scale preventive formulation shall be prepared.

(b) Flow-times for the three dilutions (starting with the least dilute solution) and 0.2M sodium chloride shall be determined after proper precautions are taken to clean the viscometer in accordance with section entitled "Cleaning of Viscometer" of ASTM D 445. The reduced specific viscosity ($\overline{\eta}$), expressed in L/g, shall be calculated using the time (t_o) obtained with the 0.2M sodium chloride solution, and flow-through times (t_c) obtained with the three diluted samples as follows:

$$\overline{\eta} = \frac{t_c/t_o - 1}{C}$$

C is the concentration of the sample active ingredients in g/L. The shear rates at the wall ($\dot{\gamma}$) of the viscometer expressed in reciprocal (sec^{-1}), shall be determined by using the calibration constants (k) of the viscometer and the flowtime, t_c as follows:

$$\dot{\gamma} = \frac{k\ (sec^{-1})}{t_c}$$

A plot of $\overline{\eta}$ versus $\dot{\gamma}$ (four points) shall be constructed for each diluted sample, and the reduced specific viscosity is determined at a shear rate of 1000 sec^{-1} or where the curve is closest to being horizontal.

The equation for the best straight line for the points representing the reduced specific viscosities for the three concentrations (10, 15, and 20 g/L) shall be calculated using the least square method and the intrinsic viscosity shall be the y-intercept of this line."

5.2.1, line 4: Delete "smug" and substitute "snug".

5.2.1, line 6: Delete "170 x 205 mm." and substitute "100 x 123 cm (40" x 48")."

NOTE: The margin of this amendment is marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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